



Intestinal and Tissue Nematodes — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. What is the definitive host of *Ascaris lumbricoides*, and is an intermediate host required?

EASY

- A. Pig; snail intermediate host required
- B. Human; snail intermediate host required
- C. Dog; mosquito intermediate host required
- D. Human; requires two intermediate hosts
- E. Human; no intermediate host required

2. Approximately how many eggs can a single female *Ascaris lumbricoides* produce per day?

EASY

- A. ~200
- B. ~2,000
- C. ~2 million
- D. ~20
- E. ~200,000

3. What is the common name for *Trichuris trichiura*?

EASY

- A. Threadworm
- B. Roundworm
- C. Whipworm
- D. Pinworm
- E. Hookworm

4. Which drug combination is used to treat *Trichinella spiralis* infection, alongside an anthelmintic?

EASY

- A. Antihistamines only
- B. Steroids
- C. Antibiotics
- D. Antivirals
- E. Proton pump inhibitors

5. Where do adult *Trichuris trichiura* worms characteristically live in the human host?

MEDIUM

- A. Small intestine lumen only, free-floating
- B. Biliary tract
- C. Skeletal muscle
- D. Lungs
- E. Colon, with the anterior end threaded into the mucosa

6. A patient with heavy *Trichuris trichiura* infection presents with chronic bloody diarrhea and rectal prolapse. What additional finding is expected?

MEDIUM

- A. Hypereosinophilic myocarditis
- B. Periorbital oedema
- C. Biliary obstruction
- D. Iron-deficiency anemia from chronic blood loss
- E. Nasopharyngeal worm expulsion

7. How do rhabditiform larvae of *Strongyloides stercoralis* become infective filariform larvae via autoinfection?

MEDIUM

- A. Asexual transformation within the intestine itself
- B. Sexual reproduction requiring both male and female worms in soil
- C. Ingestion of a second intermediate host
- D. Transmission through mosquito bite
- E. Encystment in skeletal muscle

8. What is "larva currens" and with which organism is it associated?

MEDIUM

- A. A migrating serpiginous rash on the trunk/buttocks, seen in *Strongyloides stercoralis*
- B. A muscle biopsy finding in *Trichinella spiralis*
- C. An eye sign in *Ascaris lumbricoides*
- D. A stool finding in *Trichuris trichiura*
- E. A radiographic sign of intestinal obstruction

9. Which immune status predisposes to *Strongyloides* hyperinfection syndrome, and what is a feared complication?

MEDIUM

- A. Immunosuppression; Gram-negative meningitis and sepsis
- B. Pregnancy; biliary obstruction
- C. Childhood; rectal prolapse
- D. Malnutrition; periorbital oedema
- E. Diabetes; nasopharyngeal expulsion

10. Which soil-transmitted nematode infections are most likely to co-occur because they share the same transmission route?

HARD

- A. *Strongyloides stercoralis* and *Trichinella spiralis*
- B. *Trichinella spiralis* and *Ascaris lumbricoides*
- C. *Trichuris trichiura* and *Trichinella spiralis*
- D. *Ascaris lumbricoides* and *Trichuris trichiura*
- E. None of the nematodes discussed share transmission routes

11. What is the recommended concentration/diagnostic technique for improving detection of *Ascaris* and *Trichuris* eggs in stool?

MEDIUM

- A. Gram stain of stool
- B. Dark-field microscopy
- C. Blood culture
- D. Concentration method (e.g. formalin-ether)
- E. Skin scraping

12. Approximately how long after *Trichinella* larvae encyst in muscle do they become infective to a new host if ingested?

MEDIUM

- A. About 24 hours
- B. About 6 months
- C. About 3 weeks
- D. About 5 years
- E. Immediately upon encystment

13. Which finding on chest imaging is associated with the pulmonary phase of *Ascaris* larval migration?

MEDIUM

- A. Cavitory lesion with air-fluid level
- B. Eosinophilic pulmonary infiltrates (Löffler syndrome)
- C. Pleural effusion with empyema
- D. Calcified granuloma
- E. Ground-glass opacity from viral pneumonia

14. A patient with strongyloidiasis has up to 75% chance of showing which laboratory abnormality even without severe symptoms?

MEDIUM

- A. Mild peripheral eosinophilia or elevated IgE
- B. Marked thrombocytosis
- C. Severe hypoalbuminemia
- D. Elevated ALT/AST
- E. Macrocytic anemia

15. Which nematode is described as reproducing by a parthenogenetic manner in the human host?

MEDIUM

- A. *Ascaris lumbricoides*
- B. *Trichuris trichiura*
- C. *Trichinella spiralis*
- D. *Strongyloides stercoralis*
- E. *Enterobius vermicularis*

16. Which drug is specifically noted as safe to use for ascariasis during pregnancy?

HARD

- A. Mebendazole
- B. Pyrantel pamoate
- C. Albendazole
- D. Thiabendazole
- E. Ivermectin

17. What distinguishes the transformation of *Strongyloides* larvae in the free-living (external) cycle from the autoinfective cycle?

HARD

- A. The free-living cycle requires an insect vector
- B. The autoinfective cycle occurs only in soil
- C. The free-living cycle occurs only in muscle tissue
- D. Both cycles are identical and asexual
- E. The free-living cycle requires sexual reproduction between male and female worms in the soil

18. Why are muscles of respiration and the tongue particularly affected in trichinellosis?

HARD

- A. These muscles have the lowest blood flow, trapping larvae passively
- B. These are the only striated muscles in the body
- C. Larvae preferentially migrate to and encyst within highly vascular, metabolically active skeletal muscle, including these sites
- D. The organism has tropism exclusively for smooth muscle
- E. Larvae are filtered there by regional lymph nodes

19. A heavily immunosuppressed patient with chronic untreated strongyloidiasis develops hemorrhagic pneumonia, Gram-negative sepsis, and intestinal perforation. What is the underlying process?

HARD

- A. Löffler syndrome from a new exposure
- B. Hyperinfection syndrome from massive autoinfection
- C. *Trichinella* larval muscle migration
- D. Simple reinfection from contaminated soil
- E. Allergic bronchospasm from larva currens

20. Which combination of findings would best distinguish trichinellosis from ascariasis in a patient with myalgia and eosinophilia?

HARD

- A. Chronic bloody diarrhea and rectal prolapse
- B. Intestinal obstruction on abdominal X-ray with worms visualized
- C. Periorbital oedema and subungual haemorrhages with a history of eating undercooked pork
- D. Larvae detected in stool with no eggs seen
- E. Biliary colic with obstructive jaundice

✓ Answer key

Q	Answer	Difficulty	Why
1	E	EASY	Human is the definitive host and <i>Ascaris</i> requires no intermediate host.
2	E	EASY	A female may produce approximately 200,000 eggs per day.
3	C	EASY	<i>Trichuris trichiura</i> is known as the whipworm due to its whip-like anterior end.
4	B	EASY	Steroids are added to mebendazole/thiabendazole to control the inflammatory myositis of trichinellosis.
5	E	MEDIUM	Adult whipworms live in the colon with their thin anterior end embedded in the mucosa.
6	D	MEDIUM	Heavy whipworm infestation causes chronic mucoid/bloody diarrhea leading to iron-deficiency anemia and growth retardation.
7	A	MEDIUM	Autoinfection is the asexual transformation of rhabditiform to filariform larvae occurring within the intestine.
8	A	MEDIUM	Larva currens describes the rapidly migrating rash caused by <i>Strongyloides</i> filariform larvae in the skin.
9	A	MEDIUM	Hyperinfection results from massive autoinfection in immunosuppressed hosts and can cause Gram-negative meningitis, shock, and sepsis with high mortality.
10	D	HARD	Both are soil-transmitted via contaminated hands/food and are frequently found together.
11	D	MEDIUM	A concentration method such as formalin-ether improves egg detection yield over direct smear.
12	C	MEDIUM	Larvae become infective within about 3 weeks of coiling up in muscle cells.
13	B	MEDIUM	Löffler syndrome produces transient pulmonary infiltrates with eosinophilia during larval lung migration.
14	A	MEDIUM	Up to 75% of people with chronic strongyloidiasis have mild peripheral eosinophilia or elevated IgE.
15	D	MEDIUM	Only adult female <i>Strongyloides</i> reproduce asexually (parthenogenetically) embedded in the intestinal mucosa.
16	B	HARD	Pyrantel pamoate is highlighted as usable in pregnancy, unlike some other anthelmintics.
17	E	HARD	The external free-living cycle involves sexual reproduction in soil, while autoinfection is an asexual intestinal process.
18	C	HARD	<i>Trichinella</i> larvae encyst preferentially in highly active skeletal muscle groups, including respiratory muscles and the tongue, producing pain and dysfunction there.
19	B	HARD	Massive autoinfection in an immunosuppressed host produces hyperinfection syndrome with high mortality from disseminated disease.
20	C	HARD	Periorbital oedema, splinter haemorrhages and a pork-eating history are specific to <i>Trichinella</i> , unlike <i>Ascaris</i> ' intestinal/pulmonary presentation.